Onboarding

Interface Design Description

Eclipse Arrowhead 5.2.2 | support system | Revision 2026-11-09 | REVIEW DRAFT

Purpose and document structure

This document consolidates the system interface designs. Detailed Java service/transport IDDs are retained as separate appendices, followed by available Go and Rust interface descriptions. The current cross-language sections below precede the detailed implementation appendices. Appendix dates and release labels belong to the retained baseline, not to a newly qualified release. Existing descriptions have been assembled from the primary trees; no live interoperability or semantic-parity claim is made.

Java implementation

Primary folder: Eclipse-Arrowhead-java/ah5-Onboarding-java-spring. 4 detailed Java IDD source document(s) are included in the appendices. Their architecture, operation parameters, models, transport bindings, access rules and errors remain associated with Java. They must not be read as automatically implemented by the ports.

Go implementation

Primary folder: Eclipse-Arrowhead-go/ah5-Onboarding-go.

Onboarding - Go 5.2.2

Standalone support system, reorganized on 2026-10-28 following the Eclipse-Arrowhead-java distribution layout. Build and operate from this folder; no original Arrowhead checkout or sibling source module is required. The executable is bin/Onboarding. Entry point: ./cmd/onboarding. Module: arrowhead/ah5-support-go/onboarding. See DEPLOYMENT.md for configuration and standalone container use. Existing compatibility limits remain in the copied upstream documentation and UPSTREAM-README.md. The dedicated launcher selects only this support-system role. The formerly shared internal runtime and regression tests are copied locally, not linked from another tree. Files for common helpers and other route implementations remain in that private runtime; this packaging does not claim new functionality or Java parity. Documentation and license notices are included in this folder. Source provenance and import-only rewrites are listed in the distribution's SOURCE-MANIFEST.json. No running cloud or upstream source was changed.

Listening port

Built-in default 127.0.0.1:8435 (Arrowhead default). The shipped config.json, compose.yaml and Dockerfile use the same port on all interfaces ("listen": "0.0.0.0:8435"; 8540-8555 layout before 2026-11-09). Override with "listen" in the configuration file, SERVER_PORT=N or -server.port N; the port overrides keep the configured host. See [LISTENING-PORTS.md](../LISTENING-PORTS.md) for precedence and all systems.

Documented behavior and remaining work

System	Implemented behavior	Remaining work
Onboarding	Fingerprint/system approval, on/de/sy certificate stages, registry/identity provisioning and offboarding	Complete renewal, crash recovery, withdrawal handling and prerequisite discovery; partial compensating saga, not a distributed transaction

Detailed implementation evidence

1 native Go IDD document(s) are included in the appendices. Their protocol and storage descriptions apply to that implementation only. Current primary-folder ownership and current listening-port notes above supersede historical paths or example ports in those retained documents.

Rust implementation

Primary folder: Eclipse-Arrowhead-rust/ah5-Onboarding-rust.

Onboarding (Rust)

Arrowhead 5.2.2 Onboarding support system. Entry point src/main.rs is built as binary onboarding by the shared crate in ../ah5-support-runtime-rust (arrowhead-support-rust).

- examples/onboarding.json - example configuration (master copy: ../ah5-support-runtime-rust/examples/)

- documentation/ - SysD/SysDD/IDD/SD sources (symlink into ../ah5-support-runtime-rust/documentation/Onboarding, where the contract generator writes)

Run: cargo run -p arrowhead-support-rust --bin onboarding -- <config.json>

Listening port

Default 127.0.0.1:8435 (Arrowhead default, role onboarding). Override with "listen": "host:port" in the configuration, SERVER_PORT=N or --server.port N; the port overrides keep the configured host. See [LISTENING-PORTS.md](../LISTENING-PORTS.md) for precedence and all systems.

Documented behavior and remaining work

System	Implemented behavior	Remaining work
Onboarding	Enrollment approval administration and unique fingerprint/system assignment	Bootstrap/on/de TLS profiles and distributed provisioning/renewal/offboarding absent; calls return501

Detailed implementation evidence

1 native Rust IDD document(s) are included in the appendices. Their protocol and storage descriptions apply to that implementation only. Current primary-folder ownership and current listening-port notes above supersede historical paths or example ports in those retained documents.

Build and state ownership

Core entry points are built by arrowhead-core-rust; the ordinary support entry points are built by arrowhead-support-rust using the shared support runtime. LocalCloudManager has its own crate. Per-system documentation links may point into the shared documentation catalog. Independent identity and state configuration is still required. Read the native appendix for persistence and restart semantics.

Cross-language qualification

Identical system names or HTTP paths do not establish interoperability. For the selected implementation, verify registered service versions and metadata, method/path and payload models, null and optional-field handling, success/error bodies, authentication and per-operation authorization, timeout/retry behavior and persistence across restart. Java MQTT/AMQP descriptions apply to a port only where its own implementation evidence explicitly supports them. Missing transports must not be advertised. State conversion from Java SQL to native snapshots, secret provisioning, backup/restore and rollback require implementation-specific procedures. No data migration, deployment or cloud reconfiguration was performed for this document set.

Detailed appendices and provenance

- Java: Eclipse-Arrowhead-java/ah5-Onboarding-java-spring/documentation/v5.2.2/Onboarding-v5.2.2-generalManagement-HTTP-IDD.tex

- Java: Eclipse-Arrowhead-java/ah5-Onboarding-java-spring/documentation/v5.2.2/Onboarding-v5.2.2-onboarding-HTTP-IDD.tex

- Java: Eclipse-Arrowhead-java/ah5-Onboarding-java-spring/documentation/v5.2.2/Onboarding-v5.2.2-onboardingManagement-HTTP-IDD.tex

- Java: Eclipse-Arrowhead-java/ah5-Onboarding-java-spring/documentation/v5.2.2/Onboarding-v5.2.2-sysMonitor-AMQP-IDD.tex

- Go: Eclipse-Arrowhead-go/ah5-Onboarding-go/documentation/Onboarding/Onboarding-HTTPS-IDD.tex

- Rust: Eclipse-Arrowhead-rust/ah5-support-runtime-rust/documentation/Onboarding/Onboarding-HTTPS-IDD.tex

The package sources/ folder retains the selected LaTeX, figures, bibliography, language README/build evidence and source snapshots for newly described contracts. source-manifest.json records SHA-256 hashes of original inputs. build-report.json records whether an appendix used an existing newer PDF or was compiled from source.

generalManagement HTTP/TLS/JSON

Interface Design Description

Service ID: *"generalManagement"*

Abstract
This document describes the HTTP/1.1 protocol with TLS payload security and JSON payload encoding variant of the `generalManagement` service of the Onboarding system.

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1 Overview

This document describes the `generalManagement` service interface of the Onboarding system, which provides general management of the Onboarding system: reading its log entries and its configuration.

It's implemented using protocol, encoding as stated in below table:

Profile type	Type	Version
Transfer protocol	HTTP	1.1 [1]
Data encryption	TLS	1.3 [2], 1.2 [3]
Encoding	JSON	RFC 8259 [4]
Compression	N/A	-
Semantics	N/A	-
Ontology	N/A	-

Table 1: Communication and semantics details used for the `generalManagement` service interface

This document provides the Interface Design Description IDD to the *generalManagement – Service Description* document [5]. For further details about how this service is meant to be used, please consult that document.

The rest of this document describes how to realize the `generalManagement` service interface in detail. Both in terms of its operations (Section 2) and its information model (Section 3).

1.1 Interface registration

The service is registered in the ServiceRegistry with the interface template `generic_https`, policy `CERT_AUTH`, access address `onboarding`, port 8435 and base path `/onboarding/general/mgmt`. The `operations` property of the interface lists the operations below with their method and path relative to the base path (path parameters are placeholders such as `{id}` that the consumer fills in):

Operation	Method	Path
`get-log`	POST	`/logs`
`get-config`	GET	`/get-config`

2 Service Operations

The interfaces of the `generalManagement` service, its operations, data models and implementation are provided below. In particular, each subsection first names the HTTP method and path used to call the operation, after which it names the abstract operation from the `generalManagement` SD document, as well as input and output types.

All operations require the requester's identity, presented according to the provider's authentication policy: the client certificate (mutual TLS) and, with the `outsourced` policy, `Authorization: Bearer IDENTITY-TOKEN//<token>`; systems holding the Authentication secret key send `Bearer AUTHENTICATOR-KEY//<system>//<hash>`, and with the `declared` policy `Bearer SYSTEM//<system>` is sent. Request and response bodies are `application/json` unless stated otherwise. Errors are answered with an ErrorMessage body and the status codes `400 Bad Request` (invalid parameter), `401 Unauthorized` (missing or invalid identity), `403 Forbidden` (not permitted, e.g. by the management, authorization or blacklist filter), `404 Not Found`, `408 Request Timeout`, `423 Locked`, `500 Internal Server Error` and `503 Service Unavailable` (a system it depends on failed).

2.1 POST /onboarding/general/mgmt/logs

Operation: get-log
Input: LogRequest
Output: LogEntryListResponse

Returns a page of the system's log entries matching the filter.

A successful call returns `200 OK` with the output in the body.

```
POST /onboarding/general/mgmt/logs HTTP/1.1
Host: onboarding:8435
Authorization: Bearer IDENTITY-TOKEN//<token>
Content-Type: application/json

{
  "pagination": {
    "page": 0,
    "size": 10,
    "direction": "ASC",
    "sortField": "createdAt"
  },
  "from": "2026-10-22T12:00:00Z",
  "to": "2026-10-22T12:00:00Z",
  "severity": "INFO",
  "logger": "eu.arrowhead.serviceregistry.service.ServiceDiscoveryService"
}
```

Listing 1: An get-log invocation.

```
{
  "entries": [
    {
      "logId": "42",
      "entryDate": "2026-10-22T12:00:00Z",
      "logger": "eu.arrowhead.serviceregistry.service.ServiceDiscoveryService",
      "severity": "INFO",
      "message": "Service instance registered: TemperatureSensor|temperatureReading|1.0.0"
    }
  ],
  "count": 1
}
```

Listing 2: An get-log response.

2.2 GET /onboarding/general/mgmt/get-config?keys=*{keys}*

Operation: get-config
Output: KeyValues

Returns the effective values of the requested configuration keys. Only the public keys of the system are answered; other keys, including all secrets, are ignored.

Query parameters: *keys* (List<String>, optional). List values are repeated or comma-separated.

A successful call returns `200 OK` with the output in the body.

```
GET /onboarding/general/mgmt/get-config?keys=<keys> HTTP/1.1
Host: onboarding:8435
Authorization: Bearer IDENTITY-TOKEN//<token>
```

Listing 3: An get-config invocation.

```
{
  "map": {
    "server.port": "8443"
  }
}
```

Listing 4: An get-config response.

3 Data Models

Here, all data objects that can be part of the service calls associated with this service are listed in alphabetic order. Note that each subsection, which describes one type of object, begins with the *struct* keyword, which is meant to denote a JSON Object that must contain certain fields, or names, with values conforming to explicitly named types. Fields that are not needed may be omitted or `null` in requests; empty fields are omitted in responses. As a complement to the primary types defined in this section, there is also a list of secondary types in Section 3.7, which are used to represent things like hashes, identifiers and texts.

3.1 struct ErrorMessage

Object Field	Value Type	Description
"errorMessage"	String	Error text, if the event is an error.
"errorCode"	Number	
"exceptionType"	ExceptionType	
"origin"	String	

3.2 struct KeyValues

Configuration values of the system.

Object Field	Value Type	Description
"map"	Object<String>	The configuration keys and values; secrets are not returned.

3.3 struct LogEntry

One log entry.

Object Field	Value Type	Description
"logId"	String	Identifier of the log entry.
"entryDate"	DateTime	UTC time of the log entry.
"logger"	String	Logger (Java class) that wrote the entry; as filter, its name.
"severity"	String	Log level, e.g. `INFO`, `WARN`, `ERROR`; as filter, the minimum level.
"message"	String	Text of the log entry or status message of the job.
"exception"	String	Stack trace or exception text of the log entry, if any.

3.4 struct LogEntryListResponse

List of log entries.

Object Field	Value Type	Description
"entries"	Array<LogEntry>	Matching records of the current page.
"count"	Number	Total number of matching records (all pages).

3.5 struct LogRequest

Paged filter for the log entries stored in the system's database.

Object Field	Value Type	Description
"pagination"	Page	Paging and sorting of the result.
"from"	DateTime	Return entries from this UTC time.
"to"	DateTime	Return entries until this UTC time.
"severity"	String	Minimum log level to return.
"logger"	String	Logger (Java class) that wrote the entry; as filter, its name.

3.6 struct Page

Paging and sorting; all fields optional.

Object Field	Value Type	Description
"page"	Number	Zero-based page number.
"size"	Number	Number of records per page (at most `max.page.size`, default 1000).
"direction"	String	Sort direction, `ASC` or `DESC`.
"sortField"	String	Field to sort by.

3.7 Primitives

As all messages are encoded using the JSON format [4], the following primitive constructs, part of that standard, become available. Note that the official standard is defined in terms of parsing rules, while this list only concerns syntactic information. Furthermore, the Object and Array types are given optional generic type parameters, which are used in this document to signify when pair values or elements are expected to conform to certain types.

Type	Description
Array	A JSON Array [4]; Array<A> has elements of type A.
DateTime	JSON String with an ISO 8601 UTC timestamp, e.g. `2026-10-22T12:00:00Z`.
ExceptionType	One of `ARROWHEAD`, `INVALID_PARAMETER`, `AUTH`, `FORBIDDEN`, `DATA_NOT_FOUND`, `TIMEOUT`, `LOCKED`, `INTERNAL_SERVER_ERROR`, `EXTERNAL_SERVER_ERROR` (JSON String).
Number	JSON Number, an integer.
Object	JSON Object; Object<A> has string keys and values of type A.
String	JSON String.

4 References

[1] R. Fielding, M. Nottingham, and J. Reschke, “Http/1.1,” IETF, RFC 9112, 2022.

[2] E. Rescorla, “The transport layer security (tls) protocol version 1.3,” IETF, RFC 8446, 2018.

[3] T. Dierks and E. Rescorla, “The transport layer security (tls) protocol version 1.2,” IETF, RFC 5246, 2008.

[4] T. Bray, “The javascript object notation (json) data interchange format,” IETF, RFC 8259, 2017.

[5] J. Delsing, “generalManagement – Service Description (Onboarding),” Eclipse Arrowhead, SD 5.2.2, 2026, onboarding-v5.2.2-generalManagement-SD.pdf.

5 Revision History

5.1 Amendments

No.	Date	Version	Subject of Amendments	Author
1	2026-10-22	5.2.2	Document created from the Arrowhead LaTeX templates for the 5.2.2 Java Spring implementation, including the local extensions of this workspace.	Jerker Delsing
2	2026-10-30	5.2.2	Checked against the primary Java source tree (Eclipse-Arrowhead-java): services, operations, paths and data models unchanged.	Claude Opus 5.5 (draft)

5.2 Quality Assurance

No.	Date	Version	Approved by
1		5.2.2	

onboarding HTTP/TLS/JSON

Interface Design Description

Service ID: *"onboarding"*

Abstract
This document describes the HTTP/1.1 protocol with TLS payload security and JSON payload encoding variant of the `onboarding` service of the Onboarding system.

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1 Overview

This document describes the `onboarding` service interface of the Onboarding system, which provides the Arrowhead onboarding procedure: a device holding only an approved bootstrap certificate obtains its onboarding, device and system certificates and the identities of its systems.

It's implemented using protocol, encoding as stated in below table:

Profile type	Type	Version
Transfer protocol	HTTP	1.1 [1]
Data encryption	TLS	1.3 [2], 1.2 [3]
Encoding	JSON	RFC 8259 [4]
Compression	N/A	-
Semantics	N/A	-
Ontology	N/A	-

Table 1: Communication and semantics details used for the `onboarding` service interface

This document provides the Interface Design Description IDD to the *onboarding – Service Description* document [5]. For further details about how this service is meant to be used, please consult that document.

The rest of this document describes how to realize the `onboarding` service interface in detail. Both in terms of its operations (Section 2) and its information model (Section 3).

1.1 Interface registration

The service is registered in the ServiceRegistry with the interface template `generic_https`, policy `CERT_AUTH`, access address `onboarding`, port 8435 and base path `/onboarding`. The `operations` property of the interface lists the operations below with their method and path relative to the base path (path parameters are placeholders such as `{id}` that the consumer fills in):

Operation	Method	Path
`onboard-with-name`	POST	`/certificate/name`
`onboard-with-csr`	POST	`/certificate/csr`
`onboard-device`	POST	`/device`
`onboard-system`	POST	`/system`
`offboard-device`	POST	`/device/offboard`

2 Service Operations

The interfaces of the `onboarding` service, its operations, data models and implementation are provided below. In particular, each subsection first names the HTTP method and path used to call the operation, after which it names the abstract operation from the `onboarding` SD document, as well as input and output types.

All operations require the requester's identity, presented according to the provider's authentication policy: the client certificate (mutual TLS) and, with the `outsourced` policy, `Authorization: Bearer IDENTITY-TOKEN//<token>`; systems holding the Authentication secret key send `Bearer AUTHENTICATOR-KEY//<system>//<hash>`, and with the `declared` policy `Bearer SYSTEM//<system>` is sent. Request and response bodies are `application/json` unless stated otherwise. Errors are answered with an ErrorMessage body and the status codes `400 Bad Request` (invalid parameter), `401 Unauthorized` (missing or invalid identity), `403 Forbidden` (not permitted, e.g. by the management, authorization or blacklist filter), `404 Not Found`, `408 Request Timeout`, `423 Locked`, `500 Internal Server Error` and `503 Service Unavailable` (a system it depends on failed).

2.1 POST /onboarding/certificate/name

Operation: onboard-with-name
Input: NameRequest
Output: Response

Step 1 without CSR: Onboarding generates a key pair and returns the onboarding certificate with its private key (once).

A successful call returns `200 OK` with the output in the body.

```
POST /onboarding/certificate/name HTTP/1.1
Host: onboarding:8435
Authorization: Bearer IDENTITY-TOKEN//<token>
Content-Type: application/json

{
  "commonName": "Device1.TestCloud.Company.arrowhead.eu"
}
```

Listing 1: An onboard-with-name invocation.

```
{
  "certificateChain": [
    "string"
  ],
  "onboardingCertificate": {
    "certificate": "<Base64 DER>",
    "certificateFormat": "X.509",
    "certificateType": "SYSTEM",
    "publicKey": "string",
    "privateKey": "string",
    "expiresAt": "2026-10-22T12:00:00Z"
  },
  "services": [
    {
      "instanceId": "TemperatureSensor|temperatureReading|1.0.0",
      "provider": {
        "name": "TemperatureSensor",
        "metadata": {
          "location": "hall-1"
        },
        "version": "1.0.0",
        "addresses": [
          {}
        ],
        "device": {
```

```
          "name": "Raspberry1",
          "metadata": {
            "location": "hall-1"
          },
          "addresses": [
            {}
          ],
          "createdAt": "2026-10-22T12:00:00Z",
          "updatedAt": "2026-10-22T12:00:00Z"
        },
        "createdAt": "2026-10-22T12:00:00Z",
        "updatedAt": "2026-10-22T12:00:00Z"
      },
      "serviceDefinition": {
        "name": "temperatureReading",
        "createdAt": "2026-10-22T12:00:00Z",
        "updatedAt": "2026-10-22T12:00:00Z"
      },
      "version": "1.0.0",
      "expiresAt": "2026-10-22T12:00:00Z",
      "metadata": {
        "unit": "celsius"
      },
      "interfaces": [
        {
          "templateName": "generic_https",
          "protocol": "https",
          "policy": "CERT_AUTH",
          "properties": {
            "accessAddresses": [
              "192.168.0.10"
            ],
            "accessPort": 8080,
            "basePath": "/temperature",
            "operations": {
              "read": {
                "method": "GET",
                "path": "/read"
              }
            }
          }
        }
      ],
      "createdAt": "2026-10-22T12:00:00Z",
      "updatedAt": "2026-10-22T12:00:00Z"
    }
  ],
  "nextStep": "onboard-device"
}
```


Listing 2: An onboard-with-name response.

2.2 POST /onboarding/certificate/csr

Operation: onboard-with-csr
Input: CsrRequest
Output: Response

Step 1: signs the device's onboarding certificate (`on`) from its CSR; returns the chain and the registry descriptors. A successful call returns `200 OK` with the output in the body.

```
POST /onboarding/certificate/csr HTTP/1.1
Host: onboarding:8435
Authorization: Bearer IDENTITY-TOKEN//<token>
Content-Type: application/json

```

```
{
  "certificateSigningRequest": "<Base64 DER PKCS#10>"
}
```

Listing 3: An onboard-with-csr invocation.

```
{
  "certificateChain": [
    "string"
  ],
  "onboardingCertificate": {
    "certificate": "<Base64 DER>",
    "certificateFormat": "X.509",
    "certificateType": "SYSTEM",
    "publicKey": "string",
    "privateKey": "string",
    "expiresAt": "2026-10-22T12:00:00Z"
  },
  "services": [
    {
      "instanceId": "TemperatureSensor|temperatureReading|1.0.0",
      "provider": {
        "name": "TemperatureSensor",
        "metadata": {
          "location": "hall-1"
        },
        "version": "1.0.0",
        "addresses": [
          {}
        ],
        "device": {
          "name": "Raspberry1",
          "metadata": {
            "location": "hall-1"
          },
          "addresses": [
            {}
          ],
          "createdAt": "2026-10-22T12:00:00Z",
          "updatedAt": "2026-10-22T12:00:00Z"
        },
        "createdAt": "2026-10-22T12:00:00Z",
        "updatedAt": "2026-10-22T12:00:00Z"
      },
      "serviceDefinition": {
        "name": "temperatureReading",
        "createdAt": "2026-10-22T12:00:00Z",
        "updatedAt": "2026-10-22T12:00:00Z"
      },
      "version": "1.0.0",
      "expiresAt": "2026-10-22T12:00:00Z",
      "metadata": {
        "unit": "celsius"
      },
      "interfaces": [
        {
          "templateName": "generic_https",
          "protocol": "https",
          "policy": "CERT_AUTH",
          "properties": {
            "accessAddresses": [
              "192.168.0.10"
            ],
            "accessPort": 8080,
            "basePath": "/temperature",
            "operations": {
              "read": {
                "method": "GET",
```

```
                "path": "/read"
              }
            }
          }
        }
      }
    ],
    "createdAt": "2026-10-22T12:00:00Z",
    "updatedAt": "2026-10-22T12:00:00Z"
  }
 ],
 "nextStep": "onboard-device"
}
```

Listing 4: An onboard-with-csr response.

2.3 POST /onboarding/device

Operation: onboard-device
Input: DeviceRequest
Output: DeviceResponse

Step 2, with the onboarding certificate: signs the device certificate (`de`) and registers the device in the ServiceRegistry.

A successful call returns `200 OK` with the output in the body.

```
POST /onboarding/device HTTP/1.1
Host: onboarding:8435
Authorization: Bearer IDENTITY-TOKEN//<token>
Content-Type: application/json

{
  "certificateSigningRequest": "<Base64 DER PKCS#10>",
  "addresses": [
    "string"
  ],
  "metadata": {
    "unit": "celsius"
  }
}
```

Listing 5: An onboard-device invocation.

```
{
  "deviceName": "Raspberry1",
  "certificateChain": [
    "string"
  ],
  "deviceCertificate": {
    "certificate": "<Base64 DER>",
    "certificateFormat": "X.509",
    "certificateType": "SYSTEM",
    "publicKey": "string",
    "privateKey": "string",
    "expiresAt": "2026-10-22T12:00:00Z"
  },
  "approvedSystems": [
    "string"
  ],
  "nextStep": "onboard-device"
}
```

Listing 6: An onboard-device response.

2.4 POST /onboarding/system

Operation: onboard-system
Input: SystemRequest
Output: SystemResponse

Step 3, with the device certificate: signs the system certificate (`sy`) of an approved system and creates its Authentication identity; returns the identity password once.

A successful call returns `200 OK` with the output in the body.

```
POST /onboarding/system HTTP/1.1
Host: onboarding:8435
Authorization: Bearer IDENTITY-TOKEN//<token>
Content-Type: application/json

{
  "systemName": "TemperatureSensor",
  "certificateSigningRequest": "<Base64 DER PKCS#10>"
}
```

Listing 7: An onboard-system invocation.

```
{
  "systemName": "TemperatureSensor",
  "deviceName": "Raspberry1",
  "certificateChain": [
    "string"
  ],
  "systemCertificate": {
    "certificate": "<Base64 DER>",
    "certificateFormat": "X.509",
    "certificateType": "SYSTEM",
    "publicKey": "string",
    "privateKey": "string",
    "expiresAt": "2026-10-22T12:00:00Z"
  },
  "authenticationMethod": "PASSWORD",
  "credentials": {
    "password": "<password>"
  },
  "services": [
    {
      "instanceId": "TemperatureSensor|temperatureReading|1.0.0",
      "provider": {
        "name": "TemperatureSensor",
        "metadata": {
          "location": "hall-1"
        },
        "version": "1.0.0",
        "addresses": [
          {}
        ],
        "device": {
          "name": "Raspberry1",
          "metadata": {
            "location": "hall-1"
          },
          "addresses": [
            {}
          ],
          "createdAt": "2026-10-22T12:00:00Z",
          "updatedAt": "2026-10-22T12:00:00Z"
        },
        "createdAt": "2026-10-22T12:00:00Z",
        "updatedAt": "2026-10-22T12:00:00Z"
      },
```

```
      "serviceDefinition": {
        "name": "temperatureReading",
        "createdAt": "2026-10-22T12:00:00Z",
        "updatedAt": "2026-10-22T12:00:00Z"
      },
      "version": "1.0.0",
      "expiresAt": "2026-10-22T12:00:00Z",
      "metadata": {
        "unit": "celsius"
      },
      "interfaces": [
        {
          "templateName": "generic_https",
          "protocol": "https",
          "policy": "CERT_AUTH",
          "properties": {
            "accessAddresses": [
              "192.168.0.10"
            ],
            "accessPort": 8080,
            "basePath": "/temperature",
            "operations": {
              "read": {
                "method": "GET",
                "path": "/read"
              }
            }
          }
        }
      ],
      "createdAt": "2026-10-22T12:00:00Z",
      "updatedAt": "2026-10-22T12:00:00Z"
    }
  ],
  "nextStep": "onboard-device"
}
```


Listing 8: An onboard-system response.

2.5 POST /onboarding/device/offboard

Operation: offboard-device
Output: OffboardingReport

With the device certificate: removes the device's systems, identities and registry entries, revokes the certificates and blacklists the systems.

A successful call returns `200 OK` with the output in the body.

```
POST /onboarding/device/offboard HTTP/1.1
Host: onboarding:8435
Authorization: Bearer IDENTITY-TOKEN//<token>
```

Listing 9: An offboard-device invocation.


```
{
  "deviceName": "Raspberry1",
  "completed": [
    "string"
  ],
  "failed": [
    "string"
  ]
}
```

Listing 10: An offboard-device response.

3 Data Models

Here, all data objects that can be part of the service calls associated with this service are listed in alphabetic order. Note that each subsection, which describes one type of object, begins with the *struct* keyword, which is meant to denote a JSON Object that must contain certain fields, or names, with values conforming to explicitly named types. Fields that are not needed may be omitted or `null` in requests; empty fields are omitted in responses. As a complement to the primary types defined in this section, there is also a list of secondary types in Section 3.17, which are used to represent things like hashes, identifiers and texts.

3.1 struct Address

A network address with its type.

Object Field	Value Type	Description
"type"	String	Address type: `HOSTNAME`, `IPV4`, `IPV6` or `MAC`.
"address"	String	One network address (host name, IPv4, IPv6 or MAC address).

3.2 struct CertificateResult

An issued certificate.

Object Field	Value Type	Description
"certificate"	String	The certificate, Base64 DER.
"certificateFormat"	String	Encoding of the certificate, e.g. `X.509`.
"certificateType"	String	Certificate profile: `ON_BOARDING`, `DEVICE` or `SYSTEM`.
"publicKey"	String	Public key, Base64 X.509 SubjectPublicKey Info.
"privateKey"	String	Generated private key (only without a CSR, returned once).
"expiresAt"	DateTime	UTC expiry time; no value means no expiry.

3.3 struct CsrRequest

Onboarding with the device's own key.

Object Field	Value Type	Description
"certificateSigning Request"	String	PKCS\#10 certificate signing request, Base64 DER.

3.4 struct DeviceRequest

Request of the device certificate.

Object Field	Value Type	Description
"certificateSigning Request"	String	PKCS\#10 certificate signing request, Base64 DER.
"addresses"	Array<String>	Network addresses of the device for the ServiceRegistry.
"metadata"	Object	Device metadata for the ServiceRegistry.

3.5 struct DeviceResponse

The device certificate and the next step.

Object Field	Value Type	Description
"deviceName"	String	The onboarded device.
"certificateChain"	Array<String>	Issued certificate first, then the issuing CA (Base64 DER).
"deviceCertificate"	CertificateResult	The issued device certificate (`de`).
"approvedSystems"	Array<String>	Systems the operator approved for the device.
"nextStep"	String	The operation the caller performs next in the onboarding procedure.

3.6 struct DeviceResponseDTO

A registered device.

Object Field	Value Type	Description
"name"	String	Device name.
"metadata"	Object	Free-form metadata (JSON object).
"addresses"	Array<Address>	Network addresses with their types.
"createdAt"	DateTime	UTC time of creation.
"updatedAt"	DateTime	UTC time of the last change.

3.7 struct ErrorMessage

Object Field	Value Type	Description
"errorMessage"	String	Error text, if the event is an error.
"errorCode"	Number	
"exceptionType"	ExceptionType	
"origin"	String	

3.8 struct NameRequest

Onboarding with a generated key for a name.

Object Field	Value Type	Description
"commonName"	String	Common name of the requested certificate, `<Name>.<Cloud>.<Organization>.arrowhead.eu`.

3.9 struct OffboardingReport

Outcome of an off-boarding.

Object Field	Value Type	Description
"deviceName"	String	The off-boarded device.
"completed"	Array<String>	Off-boarding steps that succeeded.
"failed"	Array<String>	Off-boarding steps that failed.

3.10 struct Response

Result of the first onboarding step.

Object Field	Value Type	Description
"certificateChain"	Array<String>	Issued certificate first, then the issuing CA (Base64 DER).
"onboardingCertificate"	CertificateResult	The issued onboarding certificate (`on`).
"services"	Array<ServiceInstanceResponse>	Service instances of the core services the caller needs next.
"nextStep"	String	The operation the caller performs next in the onboarding procedure.

3.11 struct ServiceDefinitionResponse

A service definition.

Object Field	Value Type	Description
"name"	String	Service definition name.
"createdAt"	DateTime	UTC time of creation.
"updatedAt"	DateTime	UTC time of the last change.

3.12 struct ServiceInstanceInterfaceResponse

One interface of a service instance.

Object Field	Value Type	Description
"templateName"	String	Name of the interface template, snake_case (e.g. `generic_https`).
"protocol"	String	Protocol of the interface, e.g. `http`, `https`, `mqtt`, `mqtts`.
"policy"	String	Security policy of the interface: `NONE`, `CERT_AUTH` or one of the token policies.
"properties"	Object	Interface properties defined by the template, e.g. `accessAddresses`, `accessPort`, `basePath`, `operations`.

3.13 struct ServiceInstanceResponse

A registered service instance.

Object Field	Value Type	Description
"instanceId"	String	Service instance identifier `<System>|<serviceDefinition>|<version>`, or policy instance identifier.
"provider"	SystemResponseDTO	The provider system.
"serviceDefinition"	ServiceDefinitionResponse	The service definition.
"version"	Version	Version of the service instance.
"expiresAt"	DateTime	UTC expiry time; no value means no expiry.
"metadata"	Object	Free-form metadata (JSON object).
"interfaces"	Array<ServiceInstanceInterface Response>	Service interfaces offered by the instance.
"createdAt"	DateTime	UTC time of creation.
"updatedAt"	DateTime	UTC time of the last change.

3.14 struct SystemRequest

Request of a system certificate.

Object Field	Value Type	Description
"systemName"	String	System to onboard; must be approved for the device.
"certificateSigning Request"	String	PKCS\#10 certificate signing request, Base64 DER.

3.15 struct SystemResponse

The system certificate, identity and the next step.

Object Field	Value Type	Description
"systemName"	String	The onboarded system.
"deviceName"	String	Name of the device hosting the system.
"certificateChain"	Array<String>	Issued certificate first, then the issuing CA (Base64 DER).
"systemCertificate"	CertificateResult	The issued system certificate (`sy`).
"authenticationMethod"	String	Authentication method of the identity, `PASSWORD`.
"credentials"	Object<String>	Identity password of the new Authentication identity (returned once).
"services"	Array<ServiceInstanceResponse>	Service instances of the core services the caller needs next.
"nextStep"	String	The operation the caller performs next in the onboarding procedure.

3.16 struct SystemResponseDTO

A registered system.

Object Field	Value Type	Description
"name"	String	System name.
"metadata"	Object	Free-form metadata (JSON object).
"version"	Version	Version of the system.
"addresses"	Array<Address>	Network addresses with their types.
"device"	DeviceResponseDTO	The device hosting the system, if registered.
"createdAt"	DateTime	UTC time of creation.
"updatedAt"	DateTime	UTC time of the last change.

3.17 Primitives

As all messages are encoded using the JSON format [4], the following primitive constructs, part of that standard, become available. Note that the official standard is defined in terms of parsing rules, while this list only concerns syntactic information. Furthermore, the Object and Array types are given optional generic type parameters, which are used in this document to signify when pair values or elements are expected to conform to certain types.

Type	Description
Array	A JSON Array [4]; Array<A> has elements of type A.
DateTime	JSON String with an ISO 8601 UTC timestamp, e.g. `2026-10-22T12:00:00Z`.
ExceptionType	One of `ARROWHEAD`, `INVALID_PARAMETER`, `AUTH`, `FORBIDDEN`, `DATA_NOT_FOUND`, `TIMEOUT`, `LOCKED`, `INTERNAL_SERVER_ERROR`, `EXTERNAL_SERVER_ERROR` (JSON String).
Number	JSON Number, an integer.
Object	JSON Object; Object<A> has string keys and values of type A.
String	JSON String.
Version	JSON String, semantic version `major.minor.patch`.

4 References

[1] R. Fielding, M. Nottingham, and J. Reschke, “Http/1.1,” IETF, RFC 9112, 2022.

[2] E. Rescorla, “The transport layer security (tls) protocol version 1.3,” IETF, RFC 8446, 2018.

[3] T. Dierks and E. Rescorla, “The transport layer security (tls) protocol version 1.2,” IETF, RFC 5246, 2008.

[4] T. Bray, “The javascript object notation (json) data interchange format,” IETF, RFC 8259, 2017.

[5] J. Delsing, “onboarding – Service Description (Onboarding),” Eclipse Arrowhead, SD 5.2.2, 2026, onboarding-v5.2.2-onboarding-SD.pdf.

5 Revision History

5.1 Amendments

No.	Date	Version	Subject of Amendments	Author
1	2026-10-22	5.2.2	Document created from the Arrowhead LaTeX templates for the 5.2.2 Java Spring implementation, including the local extensions of this workspace.	Jerker Delsing
2	2026-10-30	5.2.2	Checked against the primary Java source tree (Eclipse-Arrowhead-java): services, operations, paths and data models unchanged.	Claude Opus 5.5 (draft)

5.2 Quality Assurance

No.	Date	Version	Approved by
1		5.2.2	

onboardingManagement HTTP/TLS/JSON

Interface Design Description

Service ID: *"onboardingManagement"*

Abstract
This document describes the HTTP/1.1 protocol with TLS payload security and JSON payload encoding variant of the `onboardingManagement` service of the Onboarding system.

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1 Overview

This document describes the `onboardingManagement` service interface of the Onboarding system, which provides operator approval and management of device enrollments.

It's implemented using protocol, encoding as stated in below table:

Profile type	Type	Version
Transfer protocol	HTTP	1.1 [1]
Data encryption	TLS	1.3 [2], 1.2 [3]
Encoding	JSON	RFC 8259 [4]
Compression	N/A	-
Semantics	N/A	-
Ontology	N/A	-

Table 1: Communication and semantics details used for the `onboardingManagement` service interface

This document provides the Interface Design Description IDD to the *onboardingManagement – Service Description* document [5]. For further details about how this service is meant to be used, please consult that document.

The rest of this document describes how to realize the `onboardingManagement` service interface in detail. Both in terms of its operations (Section 2) and its information model (Section 3).

1.1 Interface registration

The service is registered in the ServiceRegistry with the interface template `generic_https`, policy `CERT_AUTH`, access address `onboarding`, port 8435 and base path `/onboarding/mgmt`. The `operations` property of the interface lists the operations below with their method and path relative to the base path (path parameters are placeholders such as `{id}` that the consumer fills in):

Operation	Method	Path
`query-enrollments`	GET	`/enrollments`
`get-enrollment`	GET	`/enrollments/{deviceName}`
`approve-device`	PUT	`/enrollments/{deviceName}`
`offboard-device`	POST	`/enrollments/{deviceName}/offboard`
`remove-enrollment`	DELETE	`/enrollments/{deviceName}`

2 Service Operations

The interfaces of the `onboardingManagement` service, its operations, data models and implementation are provided below. In particular, each subsection first names the HTTP method and path used to call the operation, after which it names the abstract operation from the `onboardingManagement` SD document, as well as input and output types.

All operations require the requester's identity, presented according to the provider's authentication policy: the client certificate (mutual TLS) and, with the `outsourced` policy, `Authorization: Bearer IDENTITY-TOKEN//<token>`; systems holding the Authentication secret key send `Bearer AUTHENTICATOR-KEY//<system>//<hash>`, and with the `declared` policy `Bearer SYSTEM//<system>` is sent. Request and response bodies are `application/json` unless stated otherwise. Errors are answered with an ErrorMessage body and the status codes `400 Bad Request` (invalid parameter), `401 Unauthorized` (missing or invalid identity), `403 Forbidden` (not permitted, e.g. by the management, authorization or blacklist filter), `404 Not Found`, `408 Request Timeout`, `423 Locked`, `500 Internal Server Error` and `503 Service Unavailable` (a system it depends on failed).

2.1 GET /onboarding/mgmt/enrollments

Operation: query-enrollments
Output: EnrollmentList

Returns all enrollments.

A successful call returns `200 OK` with the output in the body.

```
GET /onboarding/mgmt/enrollments HTTP/1.1
Host: onboarding:8435
Authorization: Bearer IDENTITY-TOKEN//<token>
```

Listing 1: An query-enrollments invocation.

```
{
  "entries": [
    {
      "deviceName": "Raspberry1",
      "bootstrapFingerprint": "9f86d081884c7d659a2feaa0c55ad015a3bf4f1b2b0b822cd15d6c15b0f00a08",
      "approvedSystems": [
        "string"
      ],
      "state": "USED",
      "deviceCertificateId": 1,
      "deviceCertificateExpiresAt": "2026-10-22T12:00:00Z",
      "systems": [
        {
          "systemName": "TemperatureSensor",
          "certificateId": 1,
          "certificateExpiresAt": "2026-10-22T12:00:00Z",
          "issuedAt": "2026-10-22T12:00:00Z"
        }
      ],
      "approvedBy": "string",
      "approvedAt": "2026-10-22T12:00:00Z",
      "updatedAt": "2026-10-22T12:00:00Z",
      "lastMessage": "string"
    }
  ],
  "count": 1
}
```

Listing 2: An query-enrollments response.

2.2 GET /onboarding/mgmt/enrollments/*{deviceName}*

Operation: get-enrollment
Input: String
Output: EnrollmentView

Returns the enrollment of a device.
Path parameter: *deviceName*.
A successful call returns `200 OK` with the output in the body.

```
GET /onboarding/mgmt/enrollments/<deviceName> HTTP/1.1
Host: onboarding:8435
Authorization: Bearer IDENTITY-TOKEN//<token>
```

Listing 3: An get-enrollment invocation.

```
{
  "deviceName": "Raspberry1",
  "bootstrapFingerprint": "9f86d081884c7d659a2feaa0c55ad015a3bf4f1b2b0b822cd15d6c15b0f00a08",
  "approvedSystems": [
    "string"
  ],
  "state": "USED",
  "deviceCertificateId": 1,
  "deviceCertificateExpiresAt": "2026-10-22T12:00:00Z",
  "systems": [
    {
      "systemName": "TemperatureSensor",
      "certificateId": 1,
      "certificateExpiresAt": "2026-10-22T12:00:00Z",
      "issuedAt": "2026-10-22T12:00:00Z"
    }
  ],
  "approvedBy": "string",
  "approvedAt": "2026-10-22T12:00:00Z",
  "updatedAt": "2026-10-22T12:00:00Z",
  "lastMessage": "string"
}
```

Listing 4: An get-enrollment response.

2.3 PUT /onboarding/mgmt/enrollments/*{deviceName}*

Operation: approve-device
Input: EnrollmentRequest
Output: EnrollmentView

Approves a device: bootstrap-certificate fingerprint and the systems it may onboard.
Path parameter: *deviceName*.
A successful call returns `200 OK` with the output in the body.

```
PUT /onboarding/mgmt/enrollments/<deviceName> HTTP/1.1
Host: onboarding:8435
Authorization: Bearer IDENTITY-TOKEN//<token>
Content-Type: application/json

{
  "bootstrapFingerprint": "9f86d081884c7d659a2feaa0c55ad015a3bf4f1b2b0b822cd15d6c15b0f00a08",
  "systems": [
    "string"
  ]
}
```

Listing 5: An approve-device invocation.

```
{
  "deviceName": "Raspberry1",
  "bootstrapFingerprint": "9f86d081884c7d659a2feaa0c55ad015a3bf4f1b2b0b822cd15d6c15b0f00a08",
  "approvedSystems": [
    "string"
  ],
  "state": "USED",
  "deviceCertificateId": 1,
  "deviceCertificateExpiresAt": "2026-10-22T12:00:00Z",
  "systems": [
    {
      "systemName": "TemperatureSensor",
      "certificateId": 1,
      "certificateExpiresAt": "2026-10-22T12:00:00Z",
      "issuedAt": "2026-10-22T12:00:00Z"
    }
  ],
  "approvedBy": "string",
  "approvedAt": "2026-10-22T12:00:00Z",
  "updatedAt": "2026-10-22T12:00:00Z",
  "lastMessage": "string"
}
```

Listing 6: An approve-device response.

2.4 POST /onboarding/mgmt/enrollments/*{deviceName}*/offboard

Operation: offboard-device
Input: String
Output: OffboardingReport

Off-boards a device on behalf of the operator (as `onboarding.offboard-device`).
Path parameter: *deviceName*.
A successful call returns `200 OK` with the output in the body.

```
POST /onboarding/mgmt/enrollments/<deviceName>/offboard HTTP/1.1
Host: onboarding:8435
Authorization: Bearer IDENTITY-TOKEN//<token>
```

Listing 7: An offboard-device invocation.

```
{
  "deviceName": "Raspberry1",
  "completed": [
    "string"
  ],
  "failed": [
    "string"
  ]
}
```

Listing 8: An offboard-device response.

2.5 DELETE /onboarding/mgmt/enrollments/*{deviceName}*

Operation: remove-enrollment
Input: String
Output: OffboardingReport

Off-boards the device and removes its enrollment.
Path parameter: *deviceName*.
A successful call returns `200 OK` with the output in the body.

```
DELETE /onboarding/mgmt/enrollments/<deviceName> HTTP/1.1
Host: onboarding:8435
Authorization: Bearer IDENTITY-TOKEN//<token>
```

Listing 9: An remove-enrollment invocation.

```
{
  "deviceName": "Raspberry1",
  "completed": [
    "string"
  ],
  "failed": [
    "string"
  ]
}
```

Listing 10: An remove-enrollment response.

3 Data Models

Here, all data objects that can be part of the service calls associated with this service are listed in alphabetic order. Note that each subsection, which describes one type of object, begins with the *struct* keyword, which is meant to denote a JSON Object that must contain certain fields, or names, with values conforming to explicitly named types. Fields that are not needed may be omitted or `null` in requests; empty fields are omitted in responses. As a complement to the primary types defined in this section, there is also a list of secondary types in Section 3.7, which are used to represent things like hashes, identifiers and texts.

3.1 struct EnrollmentList

List of enrollments.

Object Field	Value Type	Description
"entries"	Array<EnrollmentView>	Matching records of the current page.
"count"	Number	Number of enrollments.

3.2 struct EnrollmentRequest

Operator approval of a device.

Object Field	Value Type	Description
"bootstrapFingerprint"	String	SHA-256 fingerprint of the approved bootstrap certificate.
"systems"	Array<String>	Systems the device may onboard.

3.3 struct EnrollmentView

An enrollment (approved device).

Object Field	Value Type	Description
"deviceName"	String	Name of the device hosting the system.
"bootstrapFingerprint"	String	SHA-256 fingerprint of the approved bootstrap certificate.
"approvedSystems"	Array<String>	Systems the operator approved for the device.
"state"	String	Enrollment state: `APPROVED`, `DEVICE_REGISTERED` or `OFFBOARDED`.
"deviceCertificateId"	Number	CertificateAuthority record of the device certificate.
"deviceCertificateExpiresAt"	DateTime	Expiry of the device certificate.
"systems"	Array<SystemEnrollmentView>	Systems onboarded for the device.
"approvedBy"	String	Operator who approved the device.
"approvedAt"	DateTime	Time of the approval.
"updatedAt"	DateTime	UTC time of the last change.
"lastMessage"	String	Last status message of the enrollment.

3.4 struct ErrorMessage

Object Field	Value Type	Description
"errorMessage"	String	Error text, if the event is an error.
"errorCode"	Number	
"exceptionType"	ExceptionType	
"origin"	String	

3.5 struct OffboardingReport

Outcome of an off-boarding.

Object Field	Value Type	Description
"deviceName"	String	The off-boarded device.
"completed"	Array<String>	Off-boarding steps that succeeded.
"failed"	Array<String>	Off-boarding steps that failed.

3.6 struct SystemEnrollmentView

A system onboarded for a device.

Object Field	Value Type	Description
"systemName"	String	Name of the system, PascalCase (e.g. `TemperatureSensor`).
"certificateId"	Number	CertificateAuthority record of the system certificate.
"certificateExpiresAt"	DateTime	Expiry of the system certificate.
"issuedAt"	DateTime	Time the certificate was issued.

3.7 Primitives

As all messages are encoded using the JSON format [4], the following primitive constructs, part of that standard, become available. Note that the official standard is defined in terms of parsing rules, while this list only concerns syntactic information. Furthermore, the Object and Array types are given optional generic type parameters, which are used in this document to signify when pair values or elements are expected to conform to certain types.

Type	Description
Array	A JSON Array [4]; Array<A> has elements of type A.
DateTime	JSON String with an ISO 8601 UTC timestamp, e.g. `2026-10-22T12:00:00Z`.
ExceptionType	One of `ARROWHEAD`, `INVALID_PARAMETER`, `AUTH`, `FORBIDDEN`, `DATA_NOT_FOUND`, `TIMEOUT`, `LOCKED`, `INTERNAL_SERVER_ERROR`, `EXTERNAL_SERVER_ERROR` (JSON String).
Number	JSON Number, an integer.
Object	JSON Object; Object<A> has string keys and values of type A.
String	JSON String.

4 References

[1] R. Fielding, M. Nottingham, and J. Reschke, “Http/1.1,” IETF, RFC 9112, 2022.

[2] E. Rescorla, “The transport layer security (tls) protocol version 1.3,” IETF, RFC 8446, 2018.

[3] T. Dierks and E. Rescorla, “The transport layer security (tls) protocol version 1.2,” IETF, RFC 5246, 2008.

[4] T. Bray, “The javascript object notation (json) data interchange format,” IETF, RFC 8259, 2017.

[5] J. Delsing, “onboardingManagement – Service Description (Onboarding),” Eclipse Arrowhead, SD 5.2.2, 2026, onboarding-v5.2.2-onboardingManagement-SD.pdf.

5 Revision History

5.1 Amendments

No.	Date	Version	Subject of Amendments	Author
1	2026-10-22	5.2.2	Document created from the Arrowhead LaTeX templates for the 5.2.2 Java Spring implementation, including the local extensions of this workspace.	Jerker Delsing
2	2026-10-30	5.2.2	Checked against the primary Java source tree (Eclipse-Arrowhead-java): services, operations, paths and data models unchanged; missing primitive definitions added.	Claude Opus 5.5 (draft)

5.2 Quality Assurance

No.	Date	Version	Approved by
1		5.2.2	

sysMonitor AMQP/TLS/JSON

Interface Design Description

Service ID: *"sysMonitor"*

Abstract
This document describes the AMQP 0-9-1 protocol with TLS payload security and JSON payload encoding variant of the `sysMonitor` service of the Onboarding system (SysMonitor-AMQP-IDD-5.2.2).

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5 Revision History **8**
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1 Overview

This document describes the AMQP binding of the native `sysMonitor` service of Onboarding.
It's implemented using protocol, encoding as stated in below table:

Profile type	Type	Version
Transfer protocol	AMQP	0-9-1 [1]
Data encryption	TLS	1.3 [2], 1.2 [3]
Encoding	JSON	RFC 8259 [4]
Compression	N/A	-
Semantics	N/A	-
Ontology	N/A	-

Table 1: Communication and semantics details used for the `sysMonitor` service interface

This document provides the Interface Design Description IDD to the *sysMonitor – Service Description* document [5].

1.1 Interface registration

With `sysmonitor.enabled=true`, Onboarding registers the service at start-up with interface template `generic_amqps` (protocol `amqps`, policy `CERT_AUTH`; `generic_amqp` without TLS) and the following properties and metadata:

```
{
  "serviceDefinitionName": "sysMonitor",
  "version": "1.0.0",
  "metadata": {
    "contract": "SysMonitor-AMQP-IDD-5.2.2",
    "schemaVersion": "1.0",
    "publisherMode": "native",
    "observationTypes": [
      "echo",
      "power",
      "status"
    ],
    "delivery": "persistent, publisher-confirmed"
  },
  "interfaces": [
    {
      "templateName": "generic_amqps",
      "protocol": "amqps",
      "policy": "CERT_AUTH",
      "properties": {
        "accessAddresses": [
          "amqp-broker"
        ],
        "accessPort": 5671,
        "exchange": "arrowhead",
        "baseRoutingKey": "TestCloud.sysmonitor.Onboarding.Onboarding",
        "operations": [
          "echo",
          "power",
          "status"
        ]
      }
    }
  ]
}
```

Listing 1: Registered service instance (TestCloud).

2 Service Operations

The publisher is the `SysMonitorPublisher` of the common library, configured with `sysmonitor.broker.address`, `sysmonitor.broker.port`, `sysmonitor.local.cloud` (default: from the system certificate), `sysmonitor.publish.interval` and `sysmonitor.ssl.*` (default: the `server.ssl` stores). The publisher connects to the broker (default `amqp-broker`, port 5671) with TLS, verifies the broker host name and authenticates with its X.509 system certificate (SASL `EXTERNAL`; the certificate CN is the broker user name). Every publication interval it publishes each observation to the topic exchange `arrowhead` with the routing key `<baseRoutingKey>.<operation>`, where the base routing key is `<Cloud>.sysmonitor.<System>.<serviceInstance>`. Messages are persistent (delivery mode 2), `application/json`, carry the observation identifier as message id, and are publisher-confirmed (timeout 5 s); a failed cycle is retried with a new connection at the next interval. The broker's topic permissions allow the system to publish only below its own base routing key.

2.1 PUBLISH `<Cloud>.sysmonitor.Onboarding.Onboarding.echo`

Operation: echo
Output: Observation

Liveness: `status` (`alive` or `degraded`), `uptimeSeconds` and a `detail` text; quality `good` or `uncertain`.

2.2 PUBLISH `<Cloud>.sysmonitor.Onboarding.Onboarding.power`

Operation: power
Output: Observation

Power supply: `supplyType=line`, `state=available`; unit `1`.

2.3 PUBLISH `<Cloud>.sysmonitor.Onboarding.Onboarding.status`

Operation: status
Output: Observation

Process state: `state=running`, `health` (`healthy`/`degraded`) and `serviceCount`, the number of registered services.

```
{
  "schemaVersion": "1.0",
  "observationId": "Onboarding-echo-1792670400000-42",
  "providerSystem": "Onboarding",
  "serviceInstance": "Onboarding",
  "observationType": "echo",
  "sourceTime": "2026-10-22T12:00:00.000Z",
  "sequence": 42,
  "value": {
    "status": "alive",
    "uptimeSeconds": 3600,
    "detail": "Onboarding is running"
  },
  "quality": "good",
  "metadata": {
    "publisherMode": "native"
  }
}
```

Listing 2: An echo observation.

3 Data Models

3.1 struct Observation

Object Field	Value Type	Description
"schemaVersion"	String	`1.0`.
"observationId"	String	`<System>-<type>-<epoch ms>-<sequence>`.
"providerSystem"	String	The publishing system.
"serviceInstance"	String	Service instance, default the system name.
"observationType"	String	`echo`, `power`, `status`.
"sourceTime"	String	ISO 8601 UTC time with milliseconds.
"sequence"	Number	Sequence number per publisher.
"value"	Object	Observed value, see the operations.
"unit"	String	`1` for `power` and `status`; absent for `echo`.
"quality"	String	`good` or `uncertain`.
"metadata"	Object	`{"publisherMode": "native"}`.

3.2 Primitives

Type	Description
Number	JSON Number, an integer.
Object	JSON Object; Object<A> has string keys and values of type A.
String	JSON String.

4 References

[1] AMQP Working Group, “Amqp 0-9-1 specification,” AMQP Working Group, Tech. Rep., 2008.

[2] E. Rescorla, “The transport layer security (tls) protocol version 1.3,” IETF, RFC 8446, 2018.

[3] T. Dierks and E. Rescorla, “The transport layer security (tls) protocol version 1.2,” IETF, RFC 5246, 2008.

[4] T. Bray, “The javascript object notation (json) data interchange format,” IETF, RFC 8259, 2017.

[5] J. Delsing, “sysMonitor – Service Description (Onboarding),” Eclipse Arrowhead, SD 5.2.2, 2026, onboarding-v5.2.2-sysMonitor-SD.pdf.

5 Revision History

5.1 Amendments

No.	Date	Version	Subject of Amendments	Author
1	2026-10-22	5.2.2	Document created from the Arrowhead LaTeX templates for the 5.2.2 Java Spring implementation, including the local extensions of this workspace.	Jerker Delsing
2	2026-10-30	5.2.2	Checked against the primary Java source tree (Eclipse-Arrowhead-java): services, operations, paths and data models unchanged.	Claude Opus 5.5 (draft)

5.2 Quality Assurance

No.	Date	Version	Approved by
1		5.2.2	

1 Interface overview

HTTPS, mandatory mutual TLS, Arrowhead identity-token header and JSON request/response documents. General management is sysop/whitelist protected; enrollment endpoints use approved certificate fingerprints and profile/serial checks instead of ordinary identity tokens. Error responses contain errorMessage, errorCode, origin and timestamp. Exact DTO and status/error parity is not certified.

1.1 HTTPS interface operations

The listener is HTTPS with JSON by default; raw configuration and CRL responses use their declared media types. A request body is capped at 1 MiB. The table records implemented routes, not a claim of complete behavioral conformance.

Method	Path	Access
GET	/onboarding/echo	certificate
GET	/onboarding/mgmt/enrollments	management
GET	/onboarding/mgmt/enrollments/{deviceName}	management
PUT	/onboarding/mgmt/enrollments/{deviceName}	management
POST	/onboarding/certificate/name	enrollment
POST	/onboarding/certificate/csr	enrollment
POST	/onboarding/device	enrollment
POST	/onboarding/system	enrollment
POST	/onboarding/device/offboard	enrollment
POST	/onboarding/mgmt/enrollments/{deviceName}/offboard	management
DELETE	/onboarding/mgmt/enrollments/{deviceName}	management
GET	/onboarding/monitor/echo	certificate
GET	/onboarding/general/mgmt/get-config	management
POST	/onboarding/general/mgmt/logs	management

1.2 Important delimitations

This implementation is a development profile, not a certified interchangeable Java replacement. All systems use private JSON snapshots rather than the Java databases. MQTT application APIs, exhaustive DTO normalization and error parity, automatic data migration and production qualification remain outstanding.

- Enrollment records use JSON, not the Java database. System/identity/registry operations form a partial compensating saga, not a distributed transaction. Complete renewal, crash recovery, withdrawal of approved systems and prerequisite-service discovery require further work.

1 Onboarding

1.1 System overview

Native Rust development profile for local Java source `Onboarding/`. One independently configured executable, identity, port and state file per role. No Java, Go or Python process implements the service. Some essential operations are explicitly unavailable.

1.2 Design and trust boundaries

Tokio/Hyper/Axum with rustls enforces mutual TLS. Certificate chain, validity, local CN/profile and startup-configured optional CRLs are checked. CRLs need process restart to refresh; without CRLs no revocation enforcement is claimed. Tokens are verified with Authentication and bound to the TLS identity. Business calls require ConsumerAuthorization; management requires a verified sysop or explicit broad whitelist. CA provisioning instead applies Java-style named certificate-principal rights. Outbound authorities are pinned to expected CN/profile before credentials are sent, in addition to chain, SAN and time validation.

1.3 Persistent state

Independent locked atomic JSON replaces SQL. Clone/validate/fsync/rename commits prevent partially published local transactions. Rename is the commit point; directory-sync failure is an uncertain durable commit, not a rollback. Owner-only keys/password files are separate. Network effects are not distributed transactions; interrupted starts and bridge operations require reconciliation. No automatic Java state import, clustering or deployment is delivered.

1.4 Implemented behavior

Enrollment approval administration and unique fingerprint/system assignment

1.5 Open release gates

Bootstrap/on/de TLS profiles and distributed provisioning/renewal/offboarding absent; calls return501. All systems remain development profiles, not certified interchangeable Java replacements. See `COMPATIBILITY.md` for detailed DTO fields and restrictions; the Java sources remain the authoritative contract.

1.6 Produced and consumed services

Advertised implemented HTTPS definitions: `echo`, `generalManagement`, `onboarding`, `onboardingManagement`.

Authorities consumed as applicable: Authentication, ServiceRegistry, ConsumerAuthorization and explicitly configured pinned adapter/provider origins. Only implemented operations are registered. Service versions are 1.0.0 except Gatekeeper/GateTunnel 5.2.2. No AMQP sysMonitor.observe implementation is advertised.

1.7 HTTPS operations

JSON unless raw configuration bytes, PEM CRL or echo text. Bodies: 1 MiB; remote responses: 8 MiB. Header: `Bearer IDENTITY-TOKEN//token`. Unknown routes return 404, absent dependencies 503. Unported

operations return 501 after applicable access checks and are excluded from registration. Route presence is not semantic parity.

Method	Path	Access / status
GET	/onboarding/general/mgmt/get-config	management
POST	/onboarding/general/mgmt/logs	management
GET	/onboarding/monitor/echo	certificate
GET	/onboarding/mgmt/enrollments	management
GET	/onboarding/mgmt/enrollments/{deviceName}	management
PUT	/onboarding/mgmt/enrollments/{deviceName}	management
DELETE	/onboarding/mgmt/enrollments/{deviceName}	management
POST	/onboarding/mgmt/enrollments/{deviceName}/offboard	management / 501
POST	/onboarding/certificate/name	certificate / 501
POST	/onboarding/certificate/csr	certificate / 501
POST	/onboarding/device	certificate / 501
POST	/onboarding/system	certificate / 501
POST	/onboarding/device/offboard	certificate / 501
GET	/onboarding/echo	certificate

Error envelope: errorMessage, errorCode, exceptionType, origin and timestamp. Full Java status, DTO and validation equivalence requires differential testing.

2 References and qualification

See contracts.json for local Java SHA256 hashes/mappings, COMPATIBILITY.md for DTO details and gaps, and VERIFICATION.md for tests actually run. No upstream release-tag provenance, complete replacement or real-cloud interoperability is asserted.